is conditionally independent, the relevant predictive vectors can be losslessly represented in factored form (bottom left of Figure 1a, App. A.4 for details):

Theorem 2.3. *There exists a lossless linear map from this product-state sub-manifold to a factored direct-sum representation where the local predictive vectors for each subspace live in orthogonal subspaces:*

$$\boldsymbol{\eta}_{FWH}^{(x_{1:\ell})} \equiv \bigoplus_{n=1}^{N} \boldsymbol{\eta}_n^{(x_{1:\ell})} . \tag{6}$$

The map highlighted in the theorem is invertible on product states: no predictive information is lost by representing factors in orthogonal subspaces rather than the exponentially larger joint space. However, when factors are *not* conditionally independent (i.e., when observations induce correlations that cannot be factored) the predictive vector moves off the product-state manifold (Figure 1b). The factored representation then becomes lossy: projecting onto it discards information about cross-factor correlations, degrading predictions. This creates a tradeoff, as the joint representation is always faithful but expensive, and the factored representation is cheap but only lossless when factorization is valid.

The experimental results presented in Section 3 show that transformers learn factored representations when this is consistent with the training data, and learn them early in training *even when the data is not entirely consistent with this factored representation.*

2.5. The Factored World Hypothesis

The analysis above establishes that factored representations are *possible* when the data-generating process admits conditional independence, but possibility does not imply preference. A transformer with sufficient capacity could represent the full joint space regardless. The question is whether transformers *choose* to factor.

Factored World Hypothesis (FWH)

Transformers have an inductive bias toward factored representations, organizing their activations into a direct sum of low-dimensional orthogonal subspaces even when higher-dimensional representations would be more faithful.

This hypothesis makes strong empirical commitments. It is not enough for the factored representation to be *recoverable* from activations via some nonlinear transformation; we claim it is the *native* representation, accessible via linear projection. And we claim this holds even when model capacity would easily accommodate the joint alternative.

normalization in belief updating is contractive, so even beliefs that start slightly correlated will converge toward factored states.

2.6. Testable predictions

The factored world hypothesis yields the following testable predictions:

1. **Dimension.** If the data-generating process has N conditionally independent factors with latent dimensions $d_1, \ldots, d_N$, activations should concentrate in $\sum_n (d_n - 1)$ dimensions, not $(\prod_n d_n) - 1$.
2. **Orthogonality.** These dimensions should organize into N orthogonal subspaces, one per factor.
3. **Preference for factoring.** These properties should hold even when model capacity could accommodate the full joint representation, and in cases where the data generator does not perfectly factor. The factored solution is *preferred*, not merely sufficient.

3. Experiments

To test whether transformers learn representations that are factored, we train networks on sequence data emitted by generators with different types of factorizable structure. Our experiments directly test the predictions from Section 2.6. We construct data-generating processes where the ground-truth joint and factored representations are analytically computable, then probe whether trained models learn one, the other, or something in between.

Factors and token construction. In all experiments, we compose five elementary generators as factors. These processes have been used in previous work on activation geometry (Shai et al., 2024; Riechers et al., 2025), where each produces multi-dimensional fractal representations when trained independently. In each experiment, we use three **Mess3** factors (3-state HMMs, each emitting sub-tokens from a ternary alphabet $\mathcal{Z}_n = \{0, 1, 2\}$ for $n \in \{1, 2, 3\}$) and two **Bloch Walk** factors (3-dimensional GHMMs, each emitting sub-tokens from a quaternary alphabet $\mathcal{Z}_n = \{0, 1, 2, 3\}$ for $n \in \{4, 5\}$). Sub-tokens combine into a single observed token by mapping the Cartesian-product of the sub-tokens $(z_\ell^{(1)}, \ldots, z_\ell^{(5)})$ to integers: $x_\ell \in \mathcal{X} = \{\texttt{BOS}, 0, 1, \ldots, 431\}$, giving a vocabulary of size $|\mathcal{X}| = 3^3 \times 4^2 + 1 = 433$ (including BOS). Importantly, the model sees only these integer tokens and must discover any factored structure from the standard pre-training on next-token cross entropy loss.

Scope. In our synthetic generators, the predictive vector for each factor induced by a context is in one-to-one correspondence with the factor's next-subtoken distribution.

Our experiments test how transformers represent this predictive information—specifically, whether it is encoded in a joint (tensor-product) geometry or in a factorized (direct-sum) geometry. Our focus is the geometric signature of this representational choice: factored or joint.

Experimental regimes. We consider three regimes that progressively stress-test the degree to which transformer representations have an inductive bias towards factorization:

1. **Independent factors** (Section 4.1): Five processes running in parallel with transition operators $T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(z^{(n)})}$. The joint representation varies in 242 dimensions, while the factored varies in only 10.
2. **Conditionally independent factors** (Section 4.2): A dependency chain where each factor's dynamics depend on preceding factors' outputs, but tensor-product structure is preserved: $T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(z^{(n)}|z^{(1:n-1)})}$. In contrast to the previous case, here there is covariance structure amongst the states of the factors.
3. **Indecomposable generators** (Section 4.3): In this case we create a a situation where the transformer must choose between a dimensionally efficient but lossy factored representation and high dimensional but optimal joint representation. To do so we use transitions ε-close to factored form, $T^{(x)} = \varepsilon T_{\text{int}} + (1-\varepsilon) \bigotimes_{n=1}^{N} T_n^{(x)}$, where small ε keeps predictive vectors near the product-state manifold while large ε pushes them off (Figure 1b). This creates tension between dimensional efficiency and predictive fidelity.

Architecture and training. All experiments use a GPT-2 style decoder-only transformer trained with standard next-token cross-entropy loss. We train on models of capacities both above and below the dimensions required to represent the full joint geometry. The figures in the main text use models that have 4 layers, model dimension $d_{\text{model}} = 120$, and MLP dimension $d_{\text{MLP}} = 480$. We also trained on larger models, with $d_{\text{model}} = 480$, and $d_{\text{MLP}} = 1920$, with qualitatively similar results shown in Appendix D.3. We use the Adam optimizer (Kingma & Ba, 2014) with no weight decay. Training sequences have length $L = 8$. We find similar results for $L = 33$ and $L = 101$, which we show in Appendix D.3. For further training details see Appendix E. Full hyperparameters and code are available here.

4. Experimental results

Our theoretical framework makes precise predictions about the geometry of transformer activations, including the number of dimensions used, their organization into orthogonal subspaces, and the arrangement of context embeddings within each subspace. We now test these predictions systematically, beginning with the simplest case and progressively introducing complications that stress-test the degree to which transformers have an inductive bias towards factored representation.

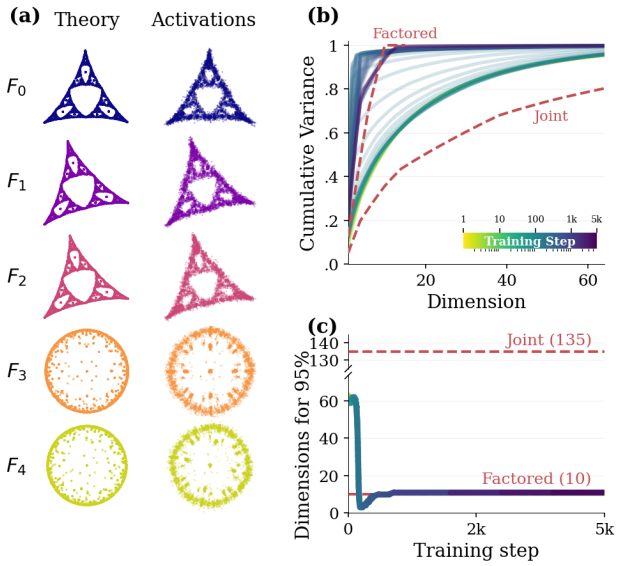

Figure 2. **Transformers discover latent factored structure.** We train a transformer on sequences generated by five independent factors (F0–F4); the model sees only tokens corresponding to the Cartesian product of factor outputs. (a) Left: ground-truth belief geometry for each factor. Right: linear regression from activations recovers each geometry with high fidelity. (b) Cumulative explained variance (CEV) over training. Dashed lines show predictions factored versus joint representations; the model converges to the factored prediction. (c) Effective dimensionality (dimensions for 95% variance) collapses to about 10 during training, far below the joint requirement. Dashed lines show dimensions required to explain 95% of the variance for the factored (10 at 95% CEV) and joint (135 at 95% CEV) predictions.

4.1. Transformers learn independent factors

We first consider the case of five factors running in complete independence, as described above. Each factor emits sub-tokens that combine into a single observed integer token. The transformer sees only these integer tokens, and nothing in the experimental setup signals to the network the existence of five underlying factors.

Each factor has a 3-dimensional latent space. The joint latent space is the tensor product of these spaces, giving dimension $3^5 = 243$. Since predictive vectors are normalized over this joint space, they vary in $243 - 1 = 242$ dimensions. In contrast, the factored representation tracks each factor's predictive vector separately. Each of these is also normalized, so each factor varies in $3 - 1 = 2$ dimensions, giving $5 \times 2 = 10$ total dimensions.

Recovering factor geometry. We first verify that the transformer has a representation of the predictive vectors at all. For each factor n, we fit a linear regression from residual stream activations to the ground-truth predictive vectors $\boldsymbol{\eta}_n^{(x_{1:\ell})}$. Figure 2(a) shows that all five factor geometries are recovered with high fidelity, with RMSE decreasing monotonically over training alongside cross-entropy loss, shown in Appendix D.1. This shows activations contain information corresponding to the latent state of each factor, but leaves open whether the model represents these factors jointly or separately.

Dimensionality of Representations The two representations make sharply different predictions about how activations are spread across the dimensions in the residual stream of the transformer, shown as dashed lines in Figure 2b. Colored lines show the cumulative explained variance (CEV) of residual stream activations over training. At initialization, the activations are spread across the residual stream dimensions,with $\sim$100 dimensions required to explain 95% of the variance (Figure 2c). Early in training, activations compress to below the number of dimensions required for the factored representations. Shortly after, the activations change their arrangement to more closely align with the prediction of the factor representation. In addition, the number of dimensions required to explain 95% of the variance plateaus at 11 dimensions, closely matching the factored prediction and ruling out the joint alternative. We verify these results hold on Transformers with d_{model} up to 480, showing a strong preference for factoring even when the residual stream has abundant capacity (Appendix D.3).

Orthogonality of factor subspaces. The factored world hypothesis predicts that each factor's information lives in its own linear subspace, and that these subspaces are mutually orthogonal. To test this, we use a "vary-one" analysis (outlined in Appendix H.1.1) to first identify candidate subspaces. We compare these subspace using two complementary orthogonality analyses to explore their relative organization in activation space (outlined in Appendix H.2).

The vary-one procedure is as follows: for each factor, we construct datasets where only that factor's sub-token sequences vary, keeping the sub-token sequences of the other four fixed. For each fixed configuration of the non-varied factors, we collect and mean-center the activations from an ensemble of realization of the varied factor. We then aggregate across many configurations and perform PCA. The top k principal components define the k-dimensional candidate subspace for that factor.

In Figure 3(a) we explore the geometric arrangement of these subspaces over training steps by tracking the effective dimensionality (at 95% variance explained) for each factor's vary-one dataset, as well as the effective dimensionality of all vary-one datasets aggregated (labeled "Union"). At initialization, the sum of individual factor dimensionalities far exceeds the union, so factors must share directions (see Appendix H.2.1). Over training, the sum falls to 14—close to the union's dimension of 12, indicating approximate but not exact orthogonality.

Figure 3(b) offers a complementary perspective by directly quantifying overlap between factor subspaces. We use a normalized subspace overlap metric that ranges from 0 (perfectly orthogonal) to 1 (fully overlapping), detailed in Appendix H.2.2. For each factor, we take the k-dimensional candidate subspace resulting from the vary-one analysis and compute pairwise overlaps across all other factor subspaces. The figure shows these pairwise overlaps at various training steps (colored lines), and the average at each step.

At initialization, factor subspaces overlap substantially regardless of k. Over training, overlaps drop markedly across all ten factor pairs, but only in the first two principal components. This is consistent with the transformer representing each factor in an approximately two-dimensional orthogonal subspace. Projecting activations onto these subspaces yields the anticipated predictive vector geometry while higher-order principal components capture relatively little variance, these results are shown in Appendix H.1.1.

Factorization begins at the embedding. Recall that the model sees only integer tokens, with nothing in the training signal indicating that each token implicitly encodes sub-tokens from five independent alphabets. Remarkably, after training, factored structure emerges at the token embedding, before any transformer blocks. The learned embedding matrix has only $\sim$12 non-negligible singular values, corresponding to the direct sum of sub-token simplices (three 2-simplices for the Mess3 factors plus two 3-simplices for the Bloch Walk factors, Figure 13). Successive principal components partition tokens by sub-token identity for each factor. This structure is absent at initialization—the model discovers the latent sub-token decomposition and assigns orthogonal subspaces to each factor at the earliest possible point (Appendix D.6).

4.2. Transformers learn conditionally independent factors

The condition for exact factoring permits dependencies among the factors, provided their joint evolution can be described by Eq. (4). These processes include all directed-acyclic graphs (DAG) of dependencies between factors (Boyd et al., 2025), with hierarchical structure among latent factors as a prominent example. To investigate whether factoring happens empirically in this broader setting, we train a transformer on data generated from a chain of three Mess3 processes and two Bloch Walk processes, similar to